Medical Insurance Cost Prediction

Predicting Healthcare Costs Using Machine Learning — A Six-Model Study

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Dataset: Medical Insurance Dataset | ~1,338 Records | 6 Features

Models Evaluated: 6 Pipelines | Final Model: Random Forest Regressor

Key Result: Test R^2 = 0.8595 | Test RMSE ≈ $4,560

A non-technical summary of methodology, findings, and business recommendations for insurance and healthcare stakeholders.

1. What Problem Were We Solving?

Health insurance pricing is one of the most consequential decisions in the insurance industry. Set premiums too low and the insurer loses money on high-cost patients; set them too high and healthy customers go elsewhere. The ability to accurately predict what a given individual will cost the insurer — before they file any claims — is enormously valuable.

This project built a machine learning system that predicts individual annual medical insurance charges based on six pieces of information routinely collected during policy sign-up: age, gender, BMI, number of dependents, smoking status, and region. The goal was to find the model that predicts charges most accurately on patients it had never seen during training.

What	Detail
Dataset size	~1,338 patient records
Information used	Age, gender, BMI, number of children, smoking status, US region
Target	Individual annual medical insurance charges (USD)
Models compared	6 progressively complex pipelines
Final model	Random Forest Regressor
Best test accuracy	$R^2 = 0.8595$ (explains 86% of cost variation); RMSE ≈ $4,560

2. Understanding the Data

Each of the 1,338 records described one insurance policyholder. The six input features cover demographics, body composition, lifestyle, and geography:

Feature	Type	What it captures
Age	Numeric	The policyholder’s age in years — older individuals generally have higher healthcare costs
Gender	Categorical	Male or female — minor cost differences observed
BMI	Numeric	Body Mass Index — a measure of weight relative to height; high BMI associated with increased health risk
Children	Numeric	Number of dependents covered under the policy
Smoking status	Categorical	Whether the policyholder smokes — turned out to be the single most powerful predictor
Region	Categorical	US residential area (Northeast, Southeast, Southwest, Northwest)

Charges ★	Target	Annual medical costs billed by the insurer — what we are trying to predict

Two important pre-modelling findings emerged from exploratory analysis: age and BMI show non-linear relationships with charges (scatter plots revealed curved rather than straight-line patterns), and the charges distribution is bimodal — two distinct groups of patients with very different cost profiles. Both of these shaped later modelling decisions.

3. Data Preparation — What We Cleaned and Why

This dataset was relatively clean, with one notable exception: BMI contained outliers at both extremes. Rather than simply deleting these patients, we applied a technique called Winsorization — capping extreme values at the statistical boundary rather than removing the records entirely.

Decision	Reasoning
Cap BMI outliers, don't delete them	Extreme BMI values represent real patients with valid (if unusual) health profiles. Removing them would bias the model toward average patients and degrade predictions for high-risk individuals — exactly the patients insurers most need to price correctly.
Keep all charge 'outliers'	169 records (~13%) appeared as outliers by standard statistical tests. Investigation revealed that 92% of these belong to smokers — a genuine high-cost sub-population, not data errors. Removing them would destroy the model's ability to price the highest-risk patients.
No missing values to handle	The dataset arrived complete. No imputation or row deletion was required for missing data.

The most important data decision in this project: the 169 apparent 'outliers' in insurance charges are not errors — 92% are smokers. They represent the highest-cost patients. Any model that removes them loses the ability to predict precisely when prediction matters most.

4. The Modelling Approach — Six Progressively Complex Pipelines

Rather than jumping to the most sophisticated model immediately, we built six pipelines of increasing complexity. Each step added a specific capability, allowing us to measure exactly what each addition contributed to prediction accuracy.

#	Model	Test R^2	What this step added
1	Linear Regression (baseline)	0.7459	Simple starting point — assumes straight-line relationships between features and cost

2	Elastic Net (regularised linear)	0.7460	Added regularisation to prevent overfitting — virtually no improvement over baseline
3	Elastic Net + Polynomial Features	0.8487	Major jump: non-linear feature interactions unlocked. R^2 rose from 0.75 to 0.85
4	Decision Tree + Polynomial Features	0.8537	Replaced linear model with tree-based splits — slight gain but some instability
5	**Random Forest + Polynomial Features ★**	**0.8595**	Ensemble of trees: highest test R^2, smallest train-test gap — WINNER
6	Random Forest + degree-2 Polynomial	Worse	Attempted to improve residuals — did not help; original Pipe 5 remained best

The single biggest accuracy gain came from adding polynomial features in Pipeline 3 — R^2 jumped from 0.75 to 0.85. This confirmed that insurance costs are driven by non-linear interactions between features, not simple straight-line relationships.

5. Why Random Forest Won

Three models exceeded 0.85 R^2 on test data. The final selection considered performance, reliability, and practical usefulness together:

Criterion	Elastic Net + Poly	Decision Tree + Poly	Random Forest ★
Test R^2	0.8487	0.8537	**0.8595 ✓**
Train-test gap	Small	Slightly larger	**Smallest (0.003)**
Overfitting risk	Low	Moderate	**Low**
Handles non-linearity	Via polynomial	Yes	**Naturally**
Interpretable features	46 coefficients—very hard	Moderate	**Feature importance scores**
Independent validation	No	No	**OOB score ✓**

The Elastic Net model, despite reasonable accuracy, expanded 6 original features into 46 polynomial coefficients — making it essentially impossible to explain to a business stakeholder. The Decision Tree was competitive but showed slightly more instability between training and test data. Random Forest won on every dimension that matters for real deployment.

6. What Drives Insurance Costs? — Feature Importance

The Random Forest model ranks which features contributed most to its predictions. The results are both statistically clear and clinically intuitive:

Rank	Feature	Importance	Business Meaning
1st	**Smoking status (yes/no)**	**~70%**	Smokers cost dramatically more to insure. This single yes/no flag explains 70% of all cost variation — more than all other features combined.
2nd	**Age**	~15%	Older policyholders have higher expected costs. A well-understood and expected relationship.
3rd	**BMI**	~10%	Higher BMI correlates with increased health risks and costs. Meaningful but secondary to smoking.
4th	**Gender, Region, Children**	~5%	Collectively minor. Region and gender have small effects; number of children has minimal individual impact.

Smoking status alone explains approximately 70% of insurance cost variability. This is not a model quirk — it reflects a real and well-documented medical reality: smokers have dramatically higher rates of cardiovascular disease, cancer, and respiratory conditions.

7. The Bimodal Cost Discovery — Why Some 'Errors' Are Not Errors

After selecting the final model, we noticed that prediction errors (residuals) were not normally distributed — they had heavier tails than expected. This is a standard diagnostic check, and a common response would be to apply mathematical transformations to 'fix' the distribution.

We tried four different approaches: log transformation, Box-Cox, Yeo-Johnson, and a more complex polynomial degree. None worked. This prompted a deeper investigation into why — which led to the most important finding of the entire project.

The Real Cause: Two Different Patient Populations

Analysis of the charges distribution revealed that the data contains two fundamentally different groups of patients:

Patient Group	Charge Pattern	Distribution
Non-smokers	Relatively lower costs	Approximately normal — well-behaved

Smokers	Significantly higher costs with a long high-cost tail	Bimodal — creating the overall non-normality in the data

Of the 169 records flagged as statistical ‘outliers’ by standard IQR analysis, 157 (92%) belonged to smokers. These are not data errors — they are real patients with genuinely high medical costs driven by smoking-related conditions.

The non-normal residuals are not a model failure — they are a reflection of a genuine structural feature in the data. The model is working correctly. The data contains two distinct sub-populations, and the Random Forest handles this bimodal structure better than any transformation could.

8. Final Model Performance

Metric	Training Set	Test Set (Unseen Data)	Interpretation
R^2 Score	0.8618	0.8595	The model explains 86% of all variation in insurance charges
RMSE (USD)	~$4,240	~$4,560	Average prediction error of ~$4,560 per patient
Train-test gap	—	0.0023	Near-zero gap — no meaningful overfitting
OOB Score	~73%	—	Independent internal validation on unseen subsets

An R^2 of 0.86 means the model captures 86% of the reason why some patients cost more than others. The remaining 14% reflects genuine unpredictability — individual health events, accidents, and conditions that cannot be forecast from six sign-up features alone.

The near-zero gap between training performance (0.8618) and test performance (0.8595) confirms that the model generalises well to new patients it has never seen — a critical requirement for any production deployment.

9. Three Business Recommendations

1 Integrate Smoking Status as a Mandatory Risk Factor in Premium Pricing

Smoking status explains approximately 70% of insurance cost variability — more than age, BMI, region, and children combined. Any actuarial or pricing model that does not explicitly capture smoking status is leaving the majority of cost signal on the table. The model demonstrates that a binary yes/no smoking flag, combined with age and BMI, is sufficient to explain 86% of cost variation. For insurers building or refreshing their premium pricing models, smoking status should be treated as a non-negotiable primary input, not an optional data field.

Strategic Impact: *Directly improves pricing accuracy for the highest-cost customer segment — smokers.*

2 Do Not Remove High-Cost ‘Outlier’ Patients From Pricing Models

Standard statistical practice would flag 169 records (~13% of this dataset) as outliers and suggest removing them before modelling. This project shows clearly why that would be wrong: 92% of those ‘outliers’ are smokers with genuinely high medical costs. Removing them would make the model accurate for average patients and useless for the highest-cost patients — precisely where pricing accuracy matters most. Any insurance analytics team applying automated outlier removal to claims data should first investigate whether the ‘outliers’ represent a specific sub-population rather than data errors.

Strategic Impact: *Protects the model’s ability to price high-risk patients accurately — where under-pricing is most costly.*

3 Use Non-Linear Models for Cost Prediction — Linear Models Leave 11% Accuracy Behind

The baseline linear regression model achieved $R^2 = 0.746$. The final Random Forest achieved $R^2 = 0.860$. This 11-percentage-point gap represents real predictive information that linear models structurally cannot capture — the non-linear interactions between age, BMI, and smoking status. In dollar terms, the Random Forest’s average prediction error (~$4,560) is meaningfully lower than what a linear model would produce. For any insurer currently using linear regression or simple rule-based pricing, upgrading to an ensemble model with polynomial feature engineering would deliver a direct improvement in pricing accuracy at relatively low implementation cost.

Strategic Impact: *11-point R^2 improvement translates directly to more accurate premium pricing and reduced adverse selection risk.*

10. Interesting Things We Found Along the Way

Polynomial features did more than any algorithm change.	Switching from linear to elastic net regularisation improved R^2 by virtually zero (0.7459 to 0.7460). Adding polynomial feature interactions in Pipeline 3 improved R^2 by 10 full points (0.7460 to 0.8487). The lesson: choosing the right feature representation matters far more than choosing the right regularisation technique when the underlying relationships are non-linear.
High accuracy on paper can hide practical uselessness.	The Elastic Net + Polynomial model achieved 0.8487 R^2 — a competitive result. But it did so using 46 polynomial coefficients derived from 6 original features. No actuary, underwriter, or executive could interpret or trust a model with 46 interacting terms. Interpretability is not a nice-to-have in financial modelling — it is a regulatory and practical requirement.
Non-normal residuals were a clue, not a problem.	Standard practice when residuals are non-normal is to transform the target variable and refit. We tried four transformations and all made things worse. The failure to fix the residuals pointed us toward the real insight: the data contains two structurally different populations. Sometimes a model diagnostic

	is telling you something about the data structure, not asking you to apply a correction.
A 1,338-row dataset is enough — if the signal is strong.	This is a relatively small dataset by modern standards. Yet the final model achieved R^2 = 0.86 with stable performance. Strong, consistent signal (smoking status driving 70% of variance) makes accurate modelling possible even with limited data. Dataset size matters less than signal quality.
Out-of-Bag scoring is an underused validation tool.	Random Forest's built-in OOB scoring provides an independent accuracy estimate without needing a separate validation set — effectively giving a third performance estimate alongside training and test scores. In small datasets where holding out a validation set reduces training data meaningfully, OOB scoring is particularly valuable.

11. Limitations & Next Steps

Known Limitations

- **Small dataset:** 1,338 records is manageable but limits the model's exposure to rare patient profiles. Retraining on a larger, real-world claims database would improve robustness and generalisability.
- **Limited features:** The six available features cover demographics and basic lifestyle. Clinical history, pre-existing conditions, occupation, and chronic disease status would likely add meaningful predictive power.
- **US-specific:** The dataset reflects US insurance market patterns. The model should not be applied to other healthcare systems without revalidation on local data.
- **Single train-test split:** Performance was evaluated on one 80/20 split. K-fold cross-validation across the full dataset would provide a more robust performance estimate.

Recommended Next Steps

- **Segment the model by smoking status:** Given that smokers and non-smokers behave as two separate populations, training separate models for each group could improve accuracy for both.
- **Add clinical history:** Integrating even basic clinical data (number of prior claims, chronic condition flags) would likely push R^2 above 0.90.
- **Deploy as a premium calculator:** The Random Forest pipeline can be serialised and integrated into an underwriting tool, providing real-time cost estimates during policy sign-up.
- **SHAP explanations for individual quotes:** Adding SHAP (SHapley Additive Explanations) analysis would allow the model to explain each individual cost prediction — essential for regulatory compliance and customer transparency.